



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
UK REACH Regulations (SI 2019/758 as amended)

Revision date 30-Sep-2024

Revision Number 1.1

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### 1.1. Product identifier

Product Name SDS Solution 20% (w/v)

Catalogue Number(s) 1610418

Pure substance/mixture Mixture

Contains Sodium lauryl sulfate

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended use Laboratory chemicals

Uses advised against No information available

### 1.3. Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Supplier

Bio-Rad Laboratories Ltd  
The Junction  
Station Road  
Watford, WD17 1ET  
UK

For further information, please contact

#### Technical Service

00800 00246 723  
Ireland: Techsupport.UK@bio-rad.com  
India: support.india@bio-rad.com  
South Africa: lsg\_techsupport\_eemea@bio-rad.com

### 1.4. Emergency telephone number

24 Hour Emergency Phone Number CHEMTREC UK: 44-870-8200418

## SECTION 2: Hazards identification

### 2.1. Classification of the substance or mixture

|                                                  |                           |
|--------------------------------------------------|---------------------------|
| Acute toxicity - Inhalation (Dusts/Mists)        | Category 4 - (H332)       |
| Skin corrosion/irritation                        | Category 2 - (H315)       |
| Serious eye damage/eye irritation                | Category 1 - (H318)       |
| Specific target organ toxicity — single exposure | Category 3 - (H335, H336) |
| Chronic aquatic toxicity                         | Category 3 - (H412)       |

### 2.2. Label elements

Contains Sodium lauryl sulfate



**Signal word**  
Danger

**Hazard statements**

H315 - Causes skin irritation

H318 - Causes serious eye damage

H332 - Harmful if inhaled

H335 + H336 - May cause respiratory irritation. May cause drowsiness or dizziness

H412 - Harmful to aquatic life with long lasting effects

**Precautionary statements**

P264 - Wash face, hands and any exposed skin thoroughly after handling

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor

P280 - Wear protective gloves/protective clothing/eye protection/face protection

**2.3. Other hazards**

No information available.

## SECTION 3: Composition/information on ingredients

**3.1 Substances**

Not applicable

**3.2 Mixtures**

| Chemical name                     | Weight-% | EC No (EU Index No) | UK REACH registration number | Classification according to GB CLP (SI 2020/1567 as amended) | Specific concentration limit (SCL) | M-Factor | M-Factor (long-term) |
|-----------------------------------|----------|---------------------|------------------------------|--------------------------------------------------------------|------------------------------------|----------|----------------------|
| Sodium lauryl sulfate<br>151-21-3 | 20 - 35  | 205-788-1           | -                            | -                                                            | -                                  | -        | -                    |

**Full text of H- and EUH-phrases: see section 16**

This product does not contain candidate substances of very high concern at a concentration  $\geq 0.1\%$  (UK REACH Article 59)

## SECTION 4: First aid measures

**4.1. Description of first aid measures**

**General advice**

Immediate medical attention is required. Show this safety data sheet to the doctor in attendance.

|                                           |                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>                         | Remove to fresh air. Get medical attention immediately if symptoms occur. IF exposed or concerned: Get medical advice/attention. If symptoms persist, call a doctor. If breathing has stopped, give artificial respiration. Get medical attention immediately.                                                   |
| <b>Eye contact</b>                        | Get immediate medical attention. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Keep eye wide open while rinsing. Do not rub affected area.                                                        |
| <b>Skin contact</b>                       | Wash off immediately with soap and plenty of water for at least 15 minutes. Get medical attention if irritation develops and persists.                                                                                                                                                                           |
| <b>Ingestion</b>                          | Rinse mouth. Never give anything by mouth to an unconscious person. Do NOT induce vomiting. Get medical attention.                                                                                                                                                                                               |
| <b>Self-protection of the first aider</b> | Avoid contact with skin, eyes or clothing. Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. Avoid breathing vapours or mists. Use personal protective equipment as required. See section 8 for more information. |

#### **4.2. Most important symptoms and effects, both acute and delayed**

|                 |                                                                                                                                                                                              |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Symptoms</b> | Burning sensation. Inhalation of high vapour concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting. Coughing and/ or wheezing. Difficulty in breathing. |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### **4.3. Indication of any immediate medical attention and special treatment needed**

|                        |                        |
|------------------------|------------------------|
| <b>Note to doctors</b> | Treat symptomatically. |
|------------------------|------------------------|

### **SECTION 5: Firefighting measures**

#### **5.1. Extinguishing media**

|                                     |                                                                                                         |
|-------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b> | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
|-------------------------------------|---------------------------------------------------------------------------------------------------------|

**Large Fire** CAUTION: Use of water spray when fighting fire may be inefficient.

|                                       |                                                                   |
|---------------------------------------|-------------------------------------------------------------------|
| <b>Unsuitable extinguishing media</b> | Do not scatter spilled material with high pressure water streams. |
|---------------------------------------|-------------------------------------------------------------------|

#### **5.2. Special hazards arising from the substance or mixture**

|                                                   |                           |
|---------------------------------------------------|---------------------------|
| <b>Specific hazards arising from the chemical</b> | No information available. |
|---------------------------------------------------|---------------------------|

#### **5.3. Advice for firefighters**

|                                                                       |                                                                                                                                    |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Special protective equipment and precautions for fire-fighters</b> | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment. |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|

### **SECTION 6: Accidental release measures**

#### **6.1. Personal precautions, protective equipment and emergency procedures**

|                             |                                                                                                                                                                                            |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal precautions</b> | Avoid contact with skin, eyes or clothing. Use personal protective equipment as required. Ensure adequate ventilation. Evacuate personnel to safe areas. Avoid breathing vapours or mists. |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                          |                                                          |
|--------------------------|----------------------------------------------------------|
| <b>Other information</b> | Refer to protective measures listed in Sections 7 and 8. |
|--------------------------|----------------------------------------------------------|

**For emergency responders** Use personal protection recommended in Section 8.

### **6.2. Environmental precautions**

**Environmental precautions** Prevent further leakage or spillage if safe to do so.

### **6.3. Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Take up mechanically, placing in appropriate containers for disposal.

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

### **6.4. Reference to other sections**

**Reference to other sections** See section 8 for more information. See section 13 for more information.

## **SECTION 7: Handling and storage**

### **7.1. Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash it before reuse. Ensure adequate ventilation. Avoid breathing vapours or mists. In case of insufficient ventilation, wear suitable respiratory equipment.

**General hygiene considerations** Avoid contact with skin, eyes or clothing. Wear suitable gloves and eye/face protection. Do not eat, drink or smoke when using this product.

### **7.2. Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Keep containers tightly closed in a dry, cool and well-ventilated place. Store locked up. Keep out of the reach of children. Store according to product and label instructions.

### **7.3. Specific end use(s)**

**Risk Management Methods (RMM)** The information required is contained in this Safety Data Sheet.

## **SECTION 8: Exposure controls/personal protection**

### **8.1. Control parameters**

**Exposure Limits** This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

**Biological occupational exposure limits** This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies.

**Derived No Effect Level (DNEL)** No information available.

**Predicted No Effect Concentration (PNEC)** No information available.

**8.2. Exposure controls**

|                                       |                                                                                                                                                                          |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Engineering controls</b>           | No information available.                                                                                                                                                |
| <b>Personal protective equipment</b>  |                                                                                                                                                                          |
| <b>Eye/face protection</b>            | Tight sealing safety goggles.                                                                                                                                            |
| <b>Hand protection</b>                | Wear suitable gloves. Impervious gloves.                                                                                                                                 |
| <b>Skin and body protection</b>       | Wear suitable protective clothing. Long sleeved clothing.                                                                                                                |
| <b>Respiratory protection</b>         | No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required. |
| <b>General hygiene considerations</b> | Avoid contact with skin, eyes or clothing. Wear suitable gloves and eye/face protection. Do not eat, drink or smoke when using this product.                             |

|                                                    |
|----------------------------------------------------|
| <b>SECTION 9: Physical and chemical properties</b> |
|----------------------------------------------------|

**9.1. Information on basic physical and chemical properties**

|                        |                          |
|------------------------|--------------------------|
| <b>Physical state</b>  | Liquid                   |
| <b>Appearance</b>      | aqueous solution         |
| <b>Colour</b>          | colourless               |
| <b>Odour</b>           | Odourless.               |
| <b>Odour threshold</b> | No information available |

| <b>Property</b>                                | <b>Values</b>            | <b>Remarks • Method</b> |
|------------------------------------------------|--------------------------|-------------------------|
| <b>Melting point / freezing point</b>          | No data available        | None known              |
| <b>Initial boiling point and boiling range</b> | > 100 °C                 |                         |
| <b>Flammability</b>                            | No data available        | None known              |
| <b>Flammability Limit in Air</b>               |                          | None known              |
| <b>Upper flammability or explosive limits</b>  | No data available        |                         |
| <b>Lower flammability or explosive limits</b>  | No data available        |                         |
| <b>Flash point</b>                             | 170 °C                   |                         |
| <b>Autoignition temperature</b>                | 248 °C                   |                         |
| <b>Decomposition temperature</b>               |                          | None known              |
| <b>pH</b>                                      |                          | None known              |
| <b>pH (as aqueous solution)</b>                | No data available        | None known              |
| <b>Kinematic viscosity</b>                     | No data available        | None known              |
| <b>Dynamic viscosity</b>                       | No data available        | None known              |
| <b>Water solubility</b>                        | Miscible in water        |                         |
| <b>Solubility(ies)</b>                         | No data available        | None known              |
| <b>Partition coefficient</b>                   | No data available        | None known              |
| <b>Vapour pressure</b>                         | No data available        | None known              |
| <b>Relative density</b>                        | No data available        | None known              |
| <b>Bulk density</b>                            | No data available        |                         |
| <b>Liquid Density</b>                          | No data available        |                         |
| <b>Relative vapour density</b>                 | No data available        | None known              |
| <b>Particle characteristics</b>                |                          |                         |
| <b>Particle Size</b>                           | No information available |                         |
| <b>Particle Size Distribution</b>              | No information available |                         |

**9.2. Other information**

**SECTION 10: Stability and reactivity****10.1. Reactivity**

**Reactivity** No information available.

**10.2. Chemical stability**

**Stability** Stable under normal conditions.

**Explosion data**

**Sensitivity to mechanical impact** None.

**Sensitivity to static discharge** None.

**10.3. Possibility of hazardous reactions**

**Possibility of hazardous reactions** None under normal processing.

**10.4. Conditions to avoid**

**Conditions to avoid** Excessive heat.

**10.5. Incompatible materials**

**Incompatible materials** Strong acids. Strong bases. Strong oxidising agents.

**10.6. Hazardous decomposition products**

**Hazardous decomposition products** None known based on information supplied.

**SECTION 11: Toxicological information****11.1. Information on toxicological effects****Information on likely routes of exposure****Product Information**

|                     |                                                                                                                                                                                              |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>   | Specific test data for the substance or mixture is not available. May cause irritation of respiratory tract. May cause drowsiness or dizziness. Harmful by inhalation (based on components). |
| <b>Eye contact</b>  | Specific test data for the substance or mixture is not available. Causes serious eye damage. May cause irreversible damage to eyes.                                                          |
| <b>Skin contact</b> | Specific test data for the substance or mixture is not available. Causes skin irritation (based on components).                                                                              |
| <b>Ingestion</b>    | Specific test data for the substance or mixture is not available. Ingestion may cause gastrointestinal irritation, nausea, vomiting and diarrhoea.                                           |

**Symptoms related to the physical, chemical and toxicological characteristics**

**Symptoms** Redness. Burning. May cause blindness. May cause redness and tearing of the eyes. Inhalation of high vapour concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting. Coughing and/ or wheezing.

**Acute toxicity**

**Numerical measures of toxicity**

The following values are calculated based on chapter 3.1 of the GHS document

ATEmix (oral) 6,440.00 mg/kg  
ATEmix (inhalation-dust/mist) 4.875 mg/l

**Unknown acute toxicity****Component Information**

| Chemical name         | Oral LD50            | Dermal LD50            | Inhalation LC50                      |
|-----------------------|----------------------|------------------------|--------------------------------------|
| Sodium lauryl sulfate | = 1288 mg/kg ( Rat ) | = 200 mg/kg ( Rabbit ) | > 3900 mg/m <sup>3</sup> ( Rat ) 1 h |

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

|                                          |                                                                                                  |
|------------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Skin corrosion/irritation</b>         | Classification based on data available for ingredients. Causes skin irritation.                  |
| <b>Serious eye damage/eye irritation</b> | Classification based on data available for ingredients. Causes burns. Causes serious eye damage. |
| <b>Respiratory or skin sensitisation</b> | No information available.                                                                        |
| <b>Germ cell mutagenicity</b>            | No information available.                                                                        |
| <b>Carcinogenicity</b>                   | No information available.                                                                        |
| <b>Reproductive toxicity</b>             | No information available.                                                                        |
| <b>STOT - single exposure</b>            | May cause respiratory irritation. May cause drowsiness or dizziness.                             |
| <b>STOT - repeated exposure</b>          | No information available.                                                                        |
| <b>Aspiration hazard</b>                 | No information available.                                                                        |
| <b>Other adverse effects</b>             | No information available.                                                                        |

**SECTION 12: Ecological information****12.1. Toxicity**

**Ecotoxicity** Harmful to aquatic life with long lasting effects.

| Chemical name         | Algae/aquatic plants                                                                              | Fish                                                                                                                                       | Toxicity to microorganisms | Crustacea                           |
|-----------------------|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-------------------------------------|
| Sodium lauryl sulfate | EC50: =53mg/L (72h, Desmodesmus subspicatus)<br>EC50: 30 - 100mg/L (96h, Desmodesmus subspicatus) | LC50: 15 - 18.9mg/L (96h, Pimephales promelas)<br>LC50: 8 - 12.5mg/L (96h, Pimephales promelas)<br>LC50: 22.1 - 22.8mg/L (96h, Pimephales) | -                          | EC50: =1.8mg/L (48h, Daphnia magna) |

|  |                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |  |
|--|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
|  | EC50: =117mg/L (96h, Pseudokirchneriella subcapitata)<br>EC50: 3.59 - 15.6mg/L (96h, Pseudokirchneriella subcapitata) | promelas)<br>LC50: 4.3 - 8.5mg/L (96h, Oncorhynchus mykiss)<br>LC50: =4.62mg/L (96h, Oncorhynchus mykiss)<br>LC50: =4.2mg/L (96h, Oncorhynchus mykiss)<br>LC50: =7.97mg/L (96h, Brachydanio rerio)<br>LC50: 9.9 - 20.1mg/L (96h, Brachydanio rerio)<br>LC50: 4.06 - 5.75mg/L (96h, Lepomis macrochirus)<br>LC50: 4.2 - 4.8mg/L (96h, Lepomis macrochirus)<br>LC50: =4.5mg/L (96h, Lepomis macrochirus)<br>LC50: 5.8 - 7.5mg/L (96h, Pimephales promelas)<br>LC50: 10.2 - 22.5mg/L (96h, Pimephales promelas)<br>LC50: 6.2 - 9.6mg/L (96h, Pimephales promelas)<br>LC50: 13.5 - 18.3mg/L (96h, Poecilia reticulata)<br>LC50: 10.8 - 16.6mg/L (96h, Poecilia reticulata)<br>LC50: =1.31mg/L (96h, Cyprinus carpio) |  |  |
|--|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|

**12.2. Persistence and degradability**

**Persistence and degradability** No information available.

**12.3. Bioaccumulative potential****Bioaccumulation****Component Information**

| Chemical name         | Partition coefficient |
|-----------------------|-----------------------|
| Sodium lauryl sulfate | 1.6                   |

**12.4. Mobility in soil**

**Mobility in soil** No information available.

**12.5. Results of PBT and vPvB assessment**

**PBT and vPvB assessment** No information available.

| Chemical name         | PBT and vPvB assessment         |
|-----------------------|---------------------------------|
| Sodium lauryl sulfate | The substance is not PBT / vPvB |

**12.6. Endocrine disrupting properties**

No information available.

## SECTION 13: Disposal considerations



**13.1. Waste treatment methods**

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

## SECTION 14: Transport information

**IATA**

|                                          |                |
|------------------------------------------|----------------|
| <b>14.1 UN number or ID number</b>       | Not regulated  |
| <b>14.2 UN proper shipping name</b>      | Not regulated  |
| <b>14.3 Transport hazard class(es)</b>   | Not regulated  |
| <b>14.4 Packing group</b>                | Not regulated  |
| <b>14.5 Environmental hazards</b>        | Not applicable |
| <b>14.6 Special precautions for user</b> |                |
| <b>Special Provisions</b>                | None           |

**IMDG**

|                                                                     |                          |
|---------------------------------------------------------------------|--------------------------|
| <b>14.1 UN number or ID number</b>                                  | Not regulated            |
| <b>14.2 UN proper shipping name</b>                                 | Not regulated            |
| <b>14.3 Transport hazard class(es)</b>                              | Not regulated            |
| <b>14.4 Packing group</b>                                           | Not regulated            |
| <b>14.5 Environmental hazards</b>                                   | Not applicable           |
| <b>14.6 Special precautions for user</b>                            |                          |
| <b>Special Provisions</b>                                           | None                     |
| <b>14.7 Maritime transport in bulk according to IMO instruments</b> | No information available |

**RID**

|                                          |                |
|------------------------------------------|----------------|
| <b>14.1 UN number or ID number</b>       | Not regulated  |
| <b>14.2 UN proper shipping name</b>      | Not regulated  |
| <b>14.3 Transport hazard class(es)</b>   | Not regulated  |
| <b>14.4 Packing group</b>                | Not regulated  |
| <b>14.5 Environmental hazards</b>        | Not applicable |
| <b>14.6 Special precautions for user</b> |                |
| <b>Special Provisions</b>                | None           |

**ADR**

|                                          |                |
|------------------------------------------|----------------|
| <b>14.1 UN number or ID number</b>       | Not regulated  |
| <b>14.2 UN proper shipping name</b>      | Not regulated  |
| <b>14.3 Transport hazard class(es)</b>   | Not regulated  |
| <b>14.4 Packing group</b>                | Not regulated  |
| <b>14.5 Environmental hazards</b>        | Not applicable |
| <b>14.6 Special precautions for user</b> |                |
| <b>Special Provisions</b>                | None           |

## SECTION 15: Regulatory information

**15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture****Authorisations and/or restrictions on use:**

This product does not contain substances subject to authorisation (UK REACH - Annex XIV). This product does not contain substances subject to restriction (UK REACH - Annex XVII).

**Persistent Organic Pollutants**

Not applicable

**Export Notification requirements**

Not applicable

**Named dangerous substances per COMAH (SI 2015/483 as amended)**

Not applicable

**The Ozone-Depleting Substances Regulations 2015**

Not applicable

**The Biocidal Products Regulations 2001 (as amended)**

Not applicable

**The Water Environment (Water Framework Directive) (England and Wales) Regulations 2017 (as amended)**

Not applicable

**Poisons and Explosive Precursors**

Not applicable

**International Inventories**

Contact supplier for inventory compliance status

**15.2. Chemical safety assessment****Chemical Safety Report**

No information available

**SECTION 16: Other information****Key or legend to abbreviations and acronyms used in the safety data sheet****Legend**

SVHC: Substances of Very High Concern for Authorisation:

**Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | Sk*  | Skin designation                 |
| +       | Sensitisers                 |      |                                  |

**Classification procedure**

|                                                                 |                       |
|-----------------------------------------------------------------|-----------------------|
| Classification according to Regulation (EC) No. 1272/2008 [CLP] | Method Used           |
| Acute oral toxicity                                             | Calculation method    |
| Acute dermal toxicity                                           | On basis of test data |
| Acute inhalation toxicity - gas                                 | Calculation method    |
| Acute inhalation toxicity - vapour                              | Calculation method    |
| Acute inhalation toxicity - dust/mist                           | Calculation method    |
| Skin corrosion/irritation                                       | Calculation method    |
| Serious eye damage/eye irritation                               | Calculation method    |
| Respiratory sensitisation                                       | Calculation method    |
| Skin sensitisation                                              | Calculation method    |
| Mutagenicity                                                    | Calculation method    |
| Carcinogenicity                                                 | Calculation method    |
| Reproductive toxicity                                           | Calculation method    |
| STOT - single exposure                                          | Calculation method    |
| STOT - repeated exposure                                        | Calculation method    |
| Acute aquatic toxicity                                          | Calculation method    |
| Chronic aquatic toxicity                                        | Calculation method    |
| Aspiration hazard                                               | Calculation method    |
| Ozone                                                           | Calculation method    |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)

U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
European Chemicals Agency (ECHA) Committee for Risk Assessment (ECHA\_RAC)  
European Chemicals Agency (ECHA) (ECHA\_API)  
Environmental Protection Agency  
Acute Exposure Guideline Level(s) (AEGL(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
National Institute of Technology and Evaluation (NITE)  
Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
U.S. National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
Organisation for Economic Co-operation and Development Screening Information Data Set  
World Health Organization

**Revision date** 30-Sep-2024

**Revision Note** Significant changes throughout SDS. Review all sections

**This SDS complies with the requirements of UK REACH Regulations SI 2019/758 (as amended)**

**Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work**

#### **Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**